Social Ties and Scientists' Productivity:

Evidence from Historical Correspondence Networks during the Quantum Revolution

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Job Practice Talk

Outline of talk

1. Motivation, RQ and Contribution
2. Preview of Findings
3. Historical Setting and Data
4. Empirical Strategy and Results
5. Robustness and Heterogeneity
6. Conclusion

Introduction

Motivation and Research Question

- How important is who you talk to for your career outcomes?
- Access to knowledge through online journals, ArXiv, Twitter, conferences, etc.
- Researchers also exchange ideas and comment on one another's work privately:
  - Not necessary to co-author and/or cite.
  - Chat over coffee, talk in the corridor, e-mails.
  - Such communication often not directly observable or measurable.
  - Who talks to whom, and how frequently?
- *Preexisting* social ties may help to promote ideas, access resources, publications

How do pre-existing social ties shape scientists' responses to a major intellectual breakthrough?

This Paper

- Setting: the emergence of quantum mechanics (QM) during 1925–1927.
- Data: inventory of scientific correspondence among 5,600 physicists, 1900–1952.
- Reconstruct pre-1925 correspondence ties with the founders of QM:
  - **Niels Bohr**, Werner Heisenberg, Max Born, etc. (the Copenhagen interpretation).
- In a DiD framework, compare post-1925 changes in scientific output:
  - Physicists *with* direct prior access to the founders.
  - Physicists *without* such prior access, but comparable on observables.
- Main outcomes:
  - Nb of publications, citations, and probability of publishing.

Preview of Findings

After 1925, scientists with preexisting correspondence ties to the founders experienced larger increases in scientific impact.

- Annual citations to previous work rise by up to 60% (average effect, ATT).
- The effects are particularly large for young scholars entering scene right before 1925.
- The estimated effects on annual publications are imprecise.
- The effect on annual citations persists during roughly 8-10 years, slowly decreasing.
- Robust to alternative correspondence-based treatment definitions.
- Robust to specifications.

Literature Review & Contribution

- **The Gap & Contribution: Pre-existing Networks as Mediators**
  - *Gap:* Lack of quantitative evidence on how **science-driven breakthroughs** trigger paradigm shifts and *who* captures their benefits.
  - *Contribution:* Shows how **pre-existing network ties** to the founders of a groundbreaking invention act as a mediator, determining which physicists capture the advantages of the new paradigm and accelerate their research output compared to outsiders.
  - *Key works:* Koschnick (2025).

- **Exogenous Shocks to the Scientific Ecosystem**
  - Political, war, or demographic shocks drastically alter scientific output, collaboration, and knowledge production.
  - *Key works:* Azoulay et al. (2010), Azoulay et al. (2019), Waldinger (2012), Moser, Voena & Waldinger (2014), Iaria et al. (2018).

- **Academic Networks & Reward Allocation**
  - Social ties, editorial influence, and competition determine *which groups* successfully pivot and capture the rewards of new research directions.
  - *Key works:* Hill & Stein (2025), Ductor (2022), Carrell, Figlio & Lusher (2024), Colussi (2018).

Historical setting

The Quantum-Mechanics Revolution, 1925–1927

- "Old quantum theory": 1900–1924.
- Three quantum physics centers: Göttingen, Munich and Copenhagen.
- 1925: Werner Heisenberg formulated an entirely new theory – Matrix Mechanics.
- 1927 Solvay Conference: formulation of the final principles of Quantum Mechanics.
  - The Copenhagen interpretation, wave-particle duality.
- Why Quantum Mechanics?
  - A (quasi-)exogenous shock to science.
  - A paradigm-shift nature: indeterminism, a-causality, abstract mathematics.
  - QM set up the research direction for the twentieth century atomic physics.
  - Why social networks?
  - Important role of mail correspondence at pre-digital times.
  - Scientists were exchanging ideas, establishing contacts and supported relationships not only via personal interaction but via letters (like us today).

Data and Treatment Definition

Data

- Catalog of scientific correspondence: The Inventory of Sources for the History of the Twentieth-Century Physics (ISHTCP) (**Wheaton 1993**)
  - Complete inventory: 5,600 physicists, all physics specialties, 1896-1950s Page examples
- Schematic biographical data on scientists (**Wheaton 1993**)
  - Specialty(ies), country(ies) of work, birth and death years. Example
- Publications and citations from the Web of Science, 1915–1945
  - Articles in leading physics, chemistry, math and general interest journals. List of journals
- Appendix details on collecting and digitizing the catalog. Data digitization

Sample Construction

- Authors published in leading journals. List of journals
- Publication years 1915–1915.
  - Robustness: 1919–1932 (after WWI, before the Nazi in GE).
- Scientist-year obs of founders, Einstein and Sommerfeld are excluded.
- Keep only scientists with first year of publication strictly before 1925.
- Web of Science authors matched to the physicists' names in the correspondence catalog.

Result:

- Balanced panel of scientist-year obs from 1915 to 1945.
- 48,329 obs: 1,559 scientists and 31 calendar years.

Baseline Treatment Definition

- Quantum mechanics founders and authors of the Copenhagen interpretation:
  - Bohr, Heisenberg, Born, Jordan, Dirac, Pauli, de Broglie and Schrödinger.
- **Treated** if exchanged any letters with any founder *prior to* 1925.
- **Untreated**: the remaining active physicists who did *not* have such correspondence.

Treated group	**Untreated group**	**Total**
154 scientists	1,405 scientists	1,559 scientists

The Founding Fathers of Quantum Mechanics



Bohr



Heisenberg



Born



Jordan



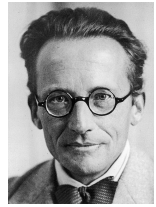
Dirac



Pauli



de Broglie



Schrödinger

Dependent Variables and Covariates

Scientist productivity measures:

- Annual citations to all previous work.
- Annual publications (adjusted by the Nb of co-authors).
- Forward citations in the next 3 and 5-years.
- Indicator equal to 1 if scientist i publishes at least once in year t.

Pre-1925 scientist covariates:

- Mean of publications, publication slope.
- Specialty and country dummies, first year of publication.
- Network centrality measures (e.g., degree).

Empirical Strategy and Results

Inverse Propensity Weighting Doubly Robust Difference-in-Difference

Baseline TWFE model

The target estimand is the average differential post-1925 change among scientists who were directly corresponding with the founders of quantum mechanics.

$$Y_{it} = \beta_0 + \beta_1(PreConnected_i \times Post1925_t) + X_i \times Post1925_t + \delta_t + \lambda_i + \epsilon_{it}$$

- Y_{it} – publication or citation outcome
- $Post1925_t$ is 1 for years $>=$ 1925, 0 otherwise
- $PreConnected_i$ is 1 if directly socially connected with the founders prior to 1925
- X_i **pre-1925 scientist-level characteristics:** specialty, working country, birth year, first year of publication, network centrality measures
- FE on Publication Year and Scientist
- Standard errors clustered at the scientist level

Baseline results

Appendix table Go

Period	1919–1932								
Outcome	Annual citations			CPA share			Cumulative citations		
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
Connected × Post-1925	3.643***	3.200***	2.660***	1.203***	0.902***	0.469*	32.115***	30.028***	20.110***
	(0.765)	(0.825)	(0.839)	(0.249)	(0.219)	(0.244)	(5.069)	(5.515)	(5.420)
Specialty × Post		✓	✓		✓	✓		✓	✓
Country × Post		✓	✓		✓	✓		✓	✓
First publication year × Post		✓	✓		✓	✓		✓	✓
Birth year × Post		✓	✓		✓	✓		✓	✓
Network characteristics × Post			✓			✓			✓
Scientist FE	✓	✓	✓	✓	✓	✓	✓	✓	✓
Year FE	✓	✓	✓	✓	✓	✓	✓	✓	✓
Obs	18,102	13,510	9,520	17,074	12,908	9,238	18,102	13,510	9,520
R^2	0.568	0.597	0.619	0.784	0.824	0.850	0.798	0.816	0.838

Notes: TWFE regressions. Standard errors clustered at the scientist level. $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.10$.

Doubly Robust IPW DiD Estimator (Callaway and Sant'Anna 2021)

Propensity score: probability of being connected to the founders before 1925, estimated using pre-treatment characteristics:

- Mean, lvl and quantiles (in extended version) of publications and citations.
- Pre-1925 prob. of publishing, Slope of publication.
- First year of publication, specialty, working country.
- Network centrality measures.
- Indirect exposure to Bohr and Einstein (number of neighbors connected to Bohr/Einstein).
- Dummies for connections (Einstein, Sommerfeld, Rutherford etc.).

Reweighting: control scientists receive weight

$$w_i = \frac{p_i}{1 - p_i},$$

where p_i is the estimated propensity score.

Goal: make the control group more comparable to pre-connected scientists in terms of observable pre-1925 characteristics.

Key Identifying Assumptions

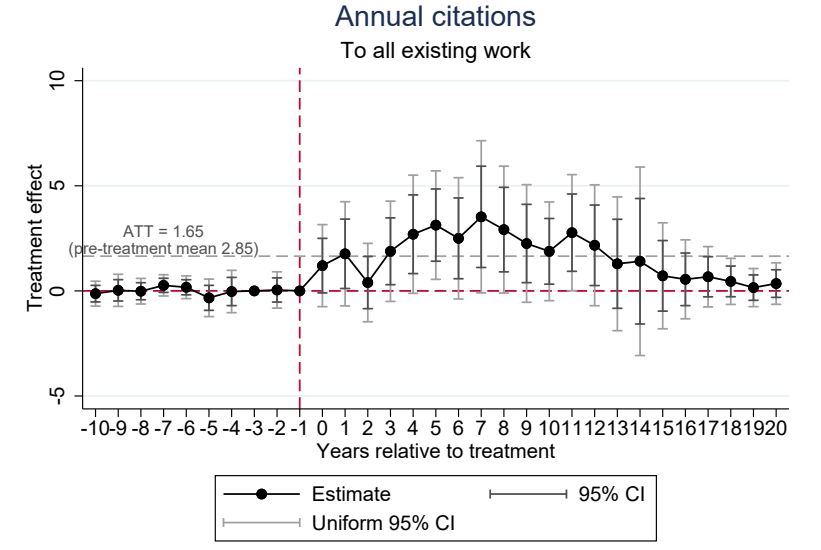
Key identifying assumption:

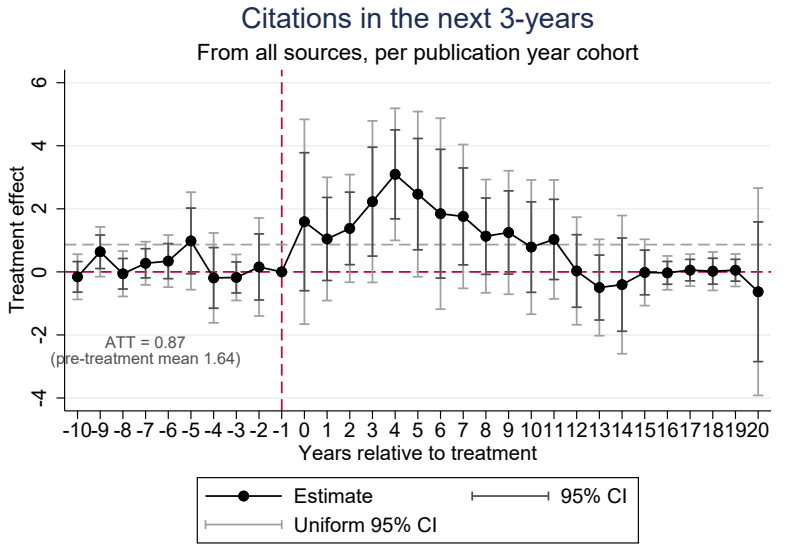
- Conditional on observed pre-1925 characteristics (balanced via IPW/Doubly Robust weights), connected and unconnected physicists would have experienced the same average changes (trends) in publications and citations after 1925, had the Quantum Mechanics breakthrough not occurred.
- Non-staggered treatment: a single shock for everyone (QM in 1925).
- **Doubly robust IPW:** combines propensity-score reweighting with an outcome-regression adjustment.
- It is robust to misspecification of either the propensity score model, or outcome regression model.
- Callaway and Sant'Anna (2021) retain cleaner causal interpretation under weaker assumptions.

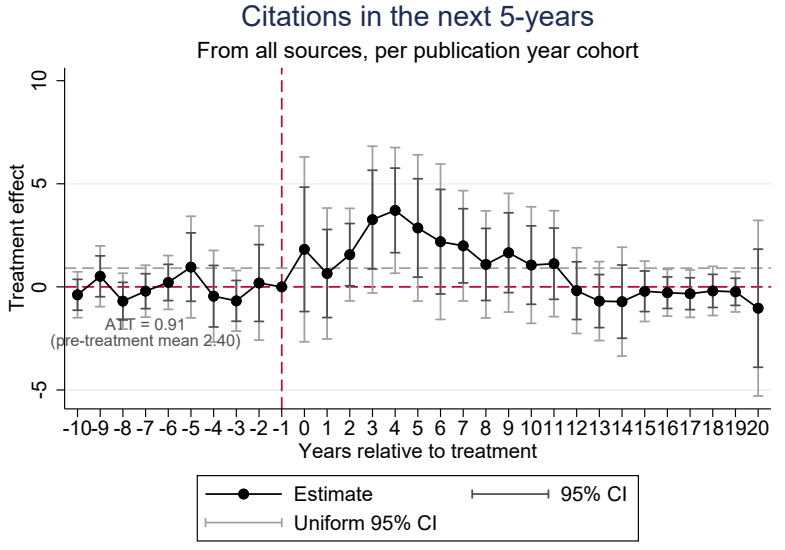
Event Study

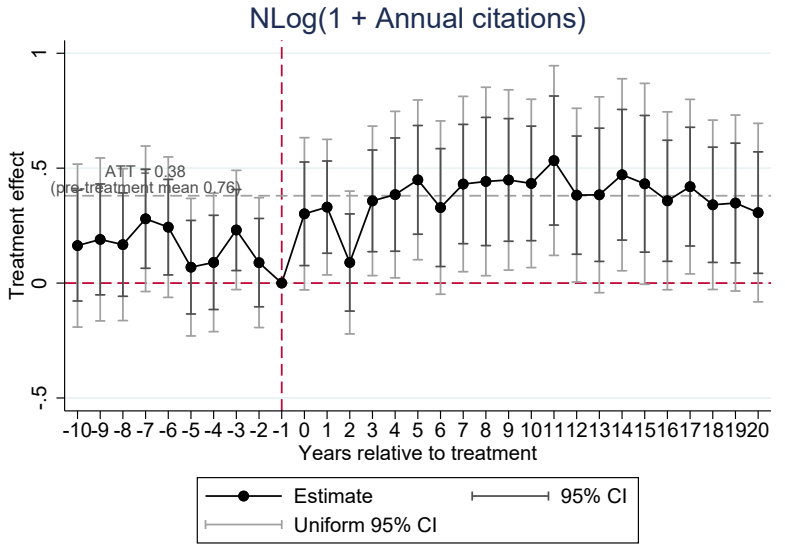
$$y_{it} = \alpha_i + \delta_t + \sum_e \beta_e \mathbf{1}\{t - 1925 = e\} \times PreConnected_i + \gamma \times X_i \times Post1925_t + \varepsilon_{it}$$

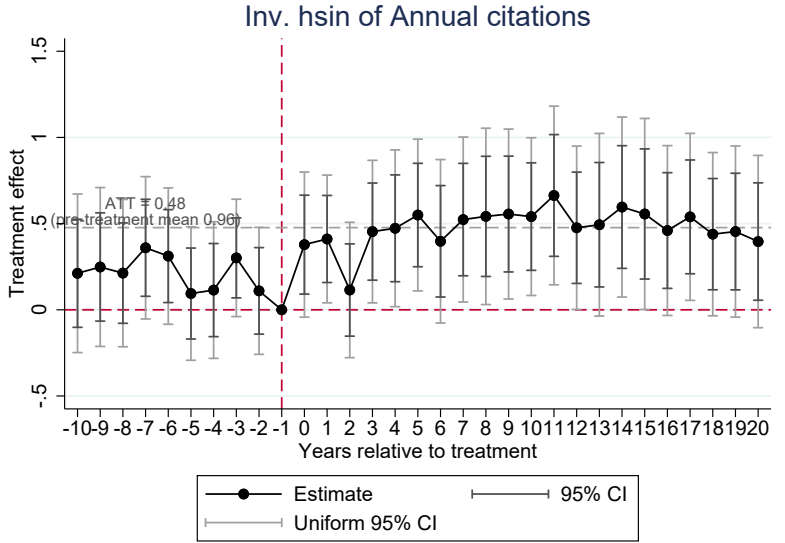
- 1925 is the treatment year for scientist i.
- Treatment occurs in the same year for each pre-connected scientist.
- $e = t - 1925$ is event time.
- β_e measures the event-time ATT at each event time, if no pre-trends holds.
- y_{it} measures changes in publication and citation outcomes:
  - Annual citations
  - Forward citations next 3 and 5-years


Annual citations
To all existing work
Treatment effect
10
5
0
-5
ATT = 1.65
(pre-treatment mean 2.85)
-10 -9 -8 -7 -6 -5 -4 -3 -2 -1 0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 18 19 20
Years relative to treatment
Estimate
95% CI
Uniform 95% CI


Citations in the next 3-years
From all sources, per publication year cohort
Treatment effect
-4
-2
0
2
4
6
ATT = 0.87
(pre-treatment mean 1.64)
-10 -9 -8 -7 -6 -5 -4 -3 -2 -1 0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 18 19 20
Years relative to treatment
Estimate
95% CI
Uniform 95% CI


Citations in the next 5-years
From all sources, per publication year cohort
Treatment effect
10
5
0
-5
ATT = 0.91
(pre-treatment mean 2.40)
-10 -9 -8 -7 -6 -5 -4 -3 -2 -1 0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 18 19 20
Years relative to treatment
Estimate
95% CI
Uniform 95% CI


NLog(1 + Annual citations)
Treatment effect
1
.5
0
-.5
ATT = 0.38
(pre-treatment mean 0.76)
-10 -9 -8 -7 -6 -5 -4 -3 -2 -1 0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 18 19 20
Years relative to treatment
Estimate
95% CI
Uniform 95% CI


Inv. hsin of Annual citations
Treatment effect
-.5
0
.5
1
1.5
ATT = 0.48
(pre-treatment mean 0.96)
-10 -9 -8 -7 -6 -5 -4 -3 -2 -1 0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 18 19 20
Years relative to treatment
Estimate
95% CI
Uniform 95% CI

Robustness and Heterogeneity

Multiple Centers for Quantum Physics

Potential exposure beyond direct connections to the eight founders

Copenhagen

- Niels Bohr
- Werner Heisenberg
- Wolfgang Pauli

Göttingen

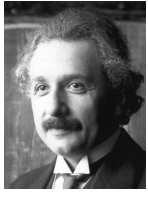
- Max Born
- Pascual Jordan
- Werner Heisenberg

Other important centers

- *Munich*: Arnold Sommerfeld
- *Berlin*: Albert Einstein
- *Zürich / Berlin*: Erwin Schrödinger
- *Paris*: Louis de Broglie
- *Cambridge*: Paul Dirac



Sommerfeld



Einstein

Re-define Treatment

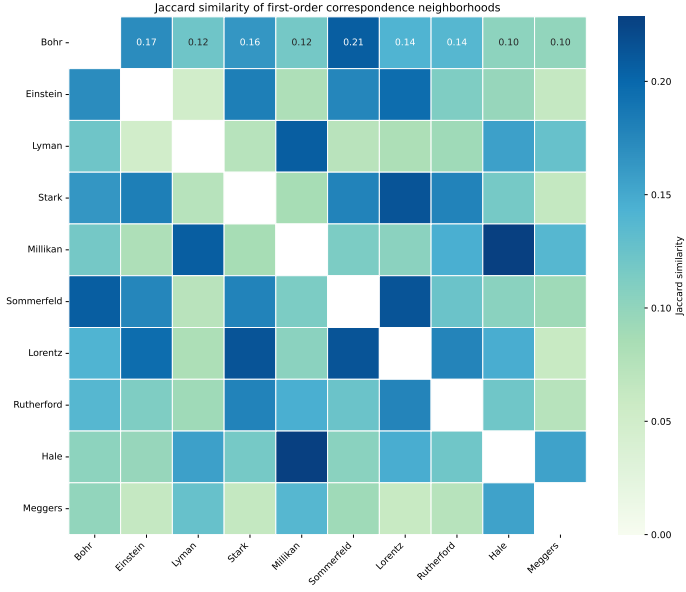
- Restrict to period 1919–1932.
- Niels Bohr – director of the *Institute of Theoretical Physics* (Copenhagen).
- **Treated** if exchanged any letters with Niels Bohr *prior to* 1925, and:
  - Did *not* have direct correspondence with Einstein (Berlin) and Sommerfeld (Munich).
- **Control:** other remaining active physicists who:
  - Did *not* have direct correspondence with Einstein and Sommerfeld.
  - Did *not* have direct scientific correspondence with the other founders.

"Cleaner" treated group	**"Cleaner" comparison group**	**Total**
92 scientists	1,040 scientists	1,132 scientists

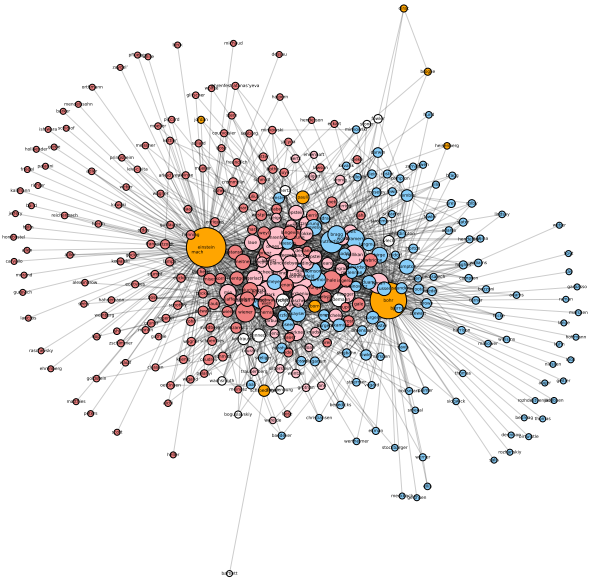
TWFE DiD Results: Bohr's Treatment and "Cleaner" Controls

Period	1919–1932					
Outcome	Annual citations			Annual publication share		
	(1)	(2)	(3)	(4)	(5)	(6)
Connected × Post-1925	1.893***	1.762**	1.664**	-0.002	-0.015	0.001
	(0.682)	(0.818)	(0.813)	(0.061)	(0.067)	(0.064)
Specialty × Post		✓	✓		✓	✓
Country × Post		✓	✓		✓	✓
Career controls × Post		✓	✓		✓	✓
Network controls × Post			✓			✓
Scientist FE	✓	✓	✓	✓	✓	✓
Year FE	✓	✓	✓	✓	✓	✓
Obs	15,848	13,020	13,020	15,848	13,020	13,020
R^2	0.551	0.569	0.570	0.340	0.366	0.366

Notes: TWFE DiD regressions. Standard errors clustered at the scientist level. $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.10$.



Note: Jaccard similarity heatmap shows the overlap among first-neighbor correspondents of selected physicists. Calculated on pre-1925 correspondence network (ISHTCP).



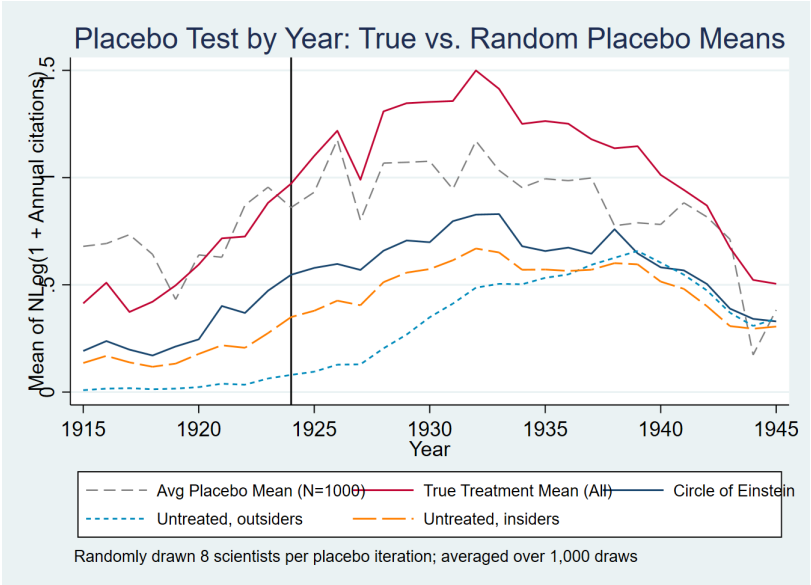
Two camps: Bohr's social circle vs. Einstein's social circle

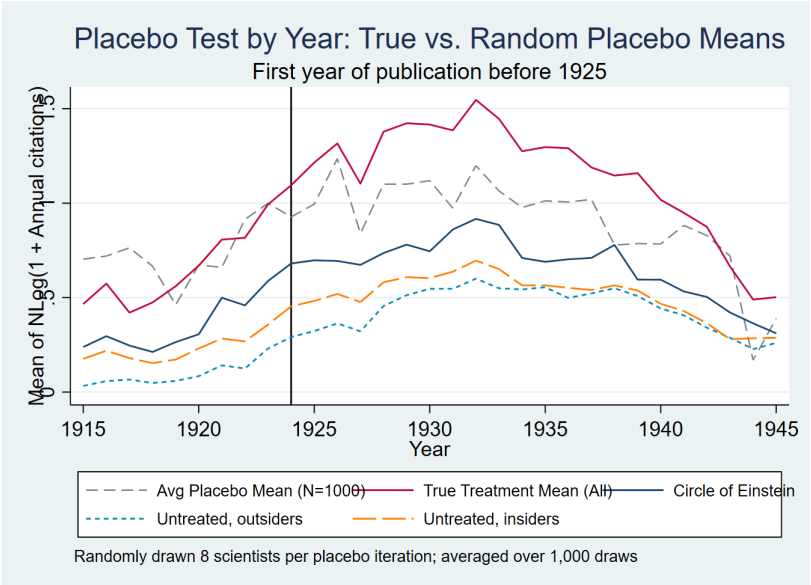
- Albert Einstein – the director of the *Kaiser Wilhelm Institute for Physics* (Berlin).
- 1922: Bohr and Einstein received the Nobel Prize in the same field – nuclear physics.
- Define treatment:
  - Treated: Bohr's correspondence circle.
  - Comparison: Einstein's correspondence circle.
- Overlapping first-neighbors ("bridges") are excluded.
- Compare post-1925 productivity change:
  - Physicists exposed to Bohr's social circle.
  - Physicists exposed to Einstein social circle.

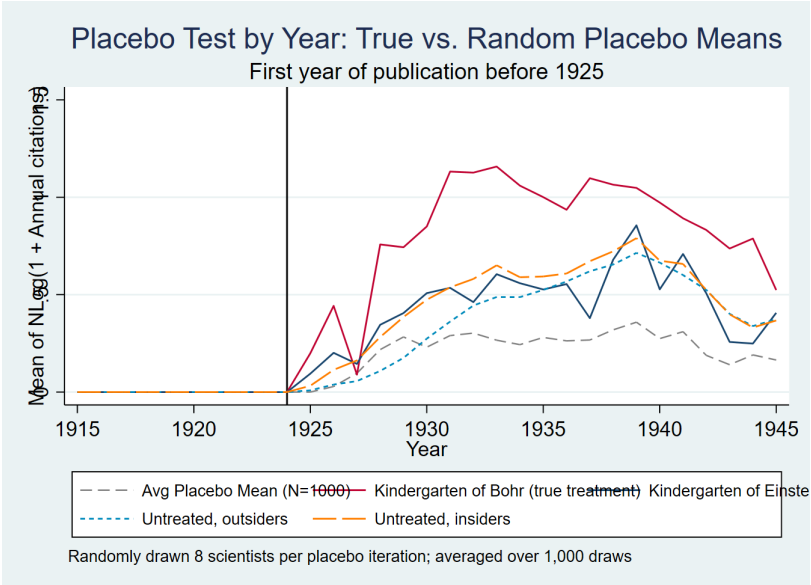
More details: Appendix details

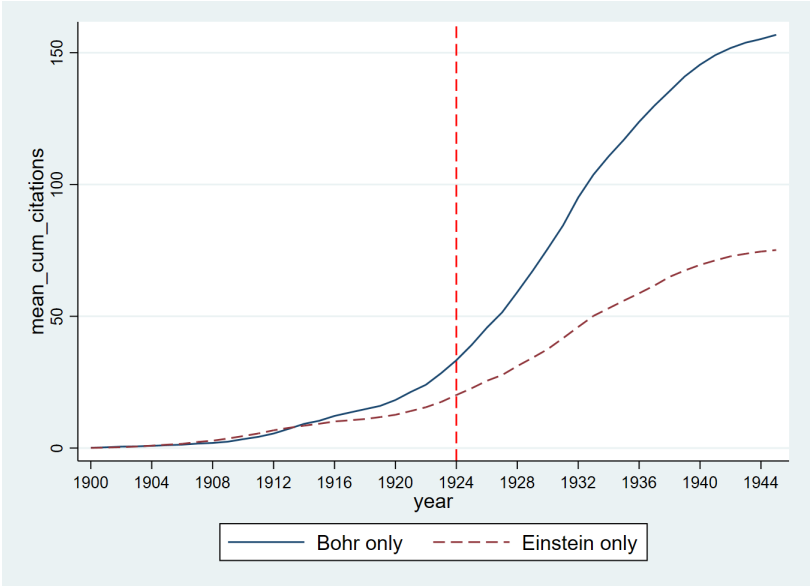
Placebo Tests

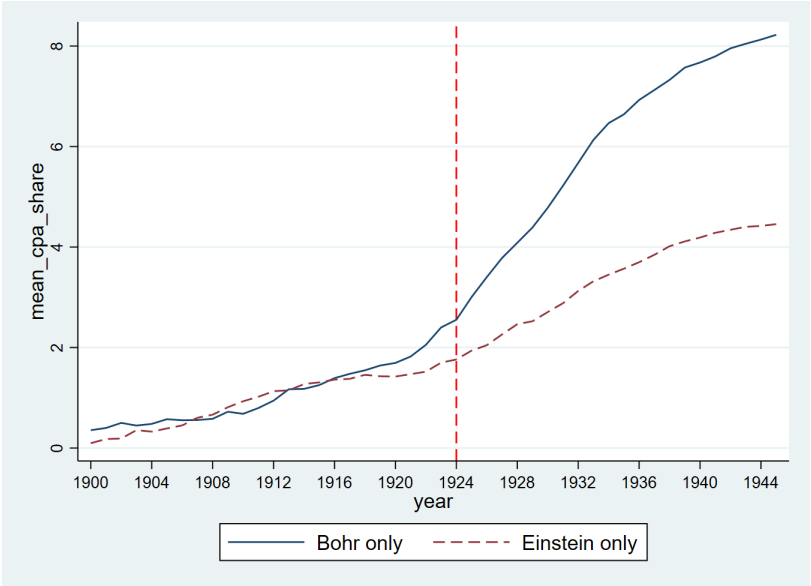
- True treatment: pre-connected with the eight QM founders (Bohr and the team).
- Placebo idea:
  - Find other prominent physicists (large hubs in the network): Sommerfeld, Einstein, Rutherford, Ehrenfest, etc.
  - Randomly pick up eight physicists and find their direct correspondents (first neighbors)
  - Calculate the average outcome over the first neighbors (citations)
  - Repeat this 1,000 times and plot the average over the repetitions
  - Compare vs. Niels Bohr and the team treatment
- Additional dimension: giant component of Bohr
  - The giant component of Bohr: correspondence links up to the fifth-neighbor distance from Bohr
  - Scientists *in* the correspondence network: **insiders**
  - Scientists *not* in the correspondence network: **outsiders**, but still **active** physicists

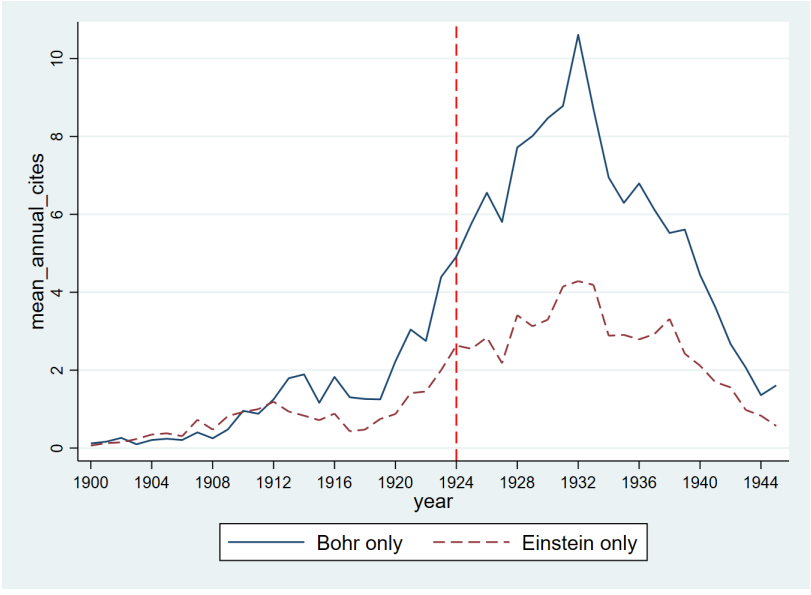

Placebo Test by Year: True vs. Random Placebo Means
Mean of NLog(1 + Annual citations)
0
.5
1
1.5
1915
1920
1925
1930
1935
1940
1945
Year
Avg Placebo Mean (N=1000)
True Treatment Mean (Alt)
Circle of Einstein
Untreated, outsiders
Untreated, insiders
Randomly drawn 8 scientists per placebo iteration; averaged over 1,000 draws

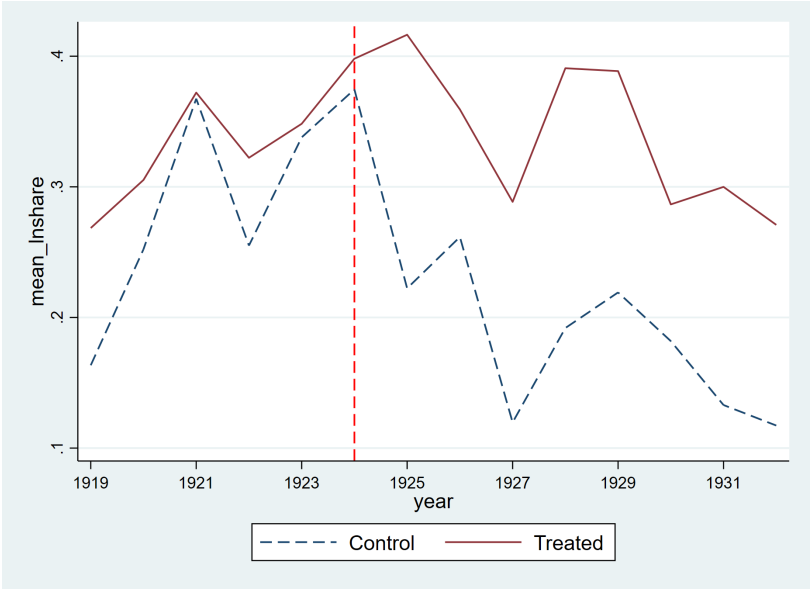

Placebo Test by Year: True vs. Random Placebo Means
First year of publication before 1925
Mean of NLog(1 + Annual citations)
0
.5
1
1.5
1915
1920
1925
1930
1935
1940
1945
Year
Avg Placebo Mean (N=1000)
True Treatment Mean (Alt)
Circle of Einstein
Untreated, outsiders
Untreated, insiders
Randomly drawn 8 scientists per placebo iteration; averaged over 1,000 draws

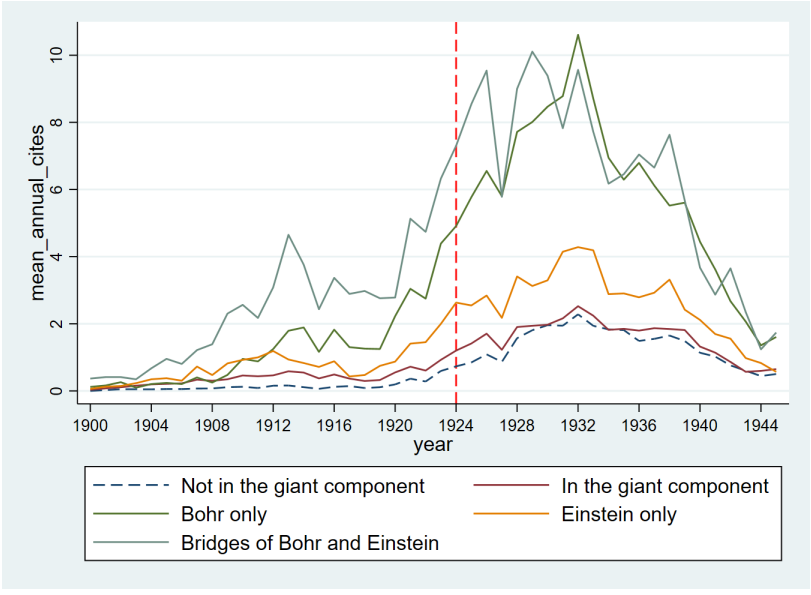

Placebo Test by Year: True vs. Random Placebo Means
First year of publication before 1925
Mean of NLog(1 + Annual citations)
0
1
2
3
1915
1920
1925
1930
1935
1940
1945
Year
Avg Placebo Mean (N=1000)
Kindergarten of Bohr (true treatment)
Kindergarten of Einste
Untreated, outsiders
Untreated, insiders
Randomly drawn 8 scientists per placebo iteration; averaged over 1,000 draws


mean_cum_citations
0
50
100
150
1900
1904
1908
1912
1916
1920
1924
1928
1932
1936
1940
1944
year
Bohr only
Einstein only


mean_cpa_share
0
2
4
6
8
1900
1904
1908
1912
1916
1920
1924
1928
1932
1936
1940
1944
year
Bohr only
Einstein only


mean_annual_cites
0
2
4
6
8
10
1900
1904
1908
1912
1916
1920
1924
1928
1932
1936
1940
1944
year
Bohr only
Einstein only


mean_lnshare
.1
.2
.3
.4
1919
1921
1923
1925
1927
1929
1931
year
Control
Treated


mean_annual_cites
0
2
4
6
8
10
1900
1904
1908
1912
1916
1920
1924
1928
1932
1936
1940
1944
year
Not in the giant component
In the giant component
Bohr only
Einstein only
Bridges of Bohr and Einstein

Conclusion

Conclusion

- **Question.** Did preexisting social ties help scientists respond to the emergence of quantum mechanics?
- **Approach.** I link historical correspondence networks to publication and citation records for 1919–1932, using pre-1925 correspondence with the founders as a proxy for direct social ties.
- **Findings.** After 1925, preconnected scientists received more annual and cumulative citations and more citations per article. They also accumulated more publications, although the effects on annual publication output are imprecisely estimated.
- **Robustness.** The main citation results persist after reweighting on pre-1925 characteristics and under alternative correspondence-based treatment definitions.
- **Interpretation.** The evidence is consistent with preexisting communication ties helping scientists engage with a new research frontier, although the correspondence data do not directly identify the transmission of specific ideas.
- Correspondence networks ties either give access to the networks. Facilitate access to the ideas, or elite networks (prestige, status and influence) or both.

Preexisting scientific ties appear to matter more for research impact than for short-run publication output

Research Agenda – Ongoing Projects on the Economics of Science and Human Capital

- Project on the diffusion of quantum mechanics ideas:
  - Track cited references: who cites the QM papers first
  - Collect letters content (sub-sample of Bohr's correspondence)
  - Can potentially prove tacit knowledge transmission through letters
  - Track physical concepts and formulas in the papers
  - Role of prior math training and age in reception of the innovation
  - Potential study of the transmission of QM to Soviet Union
- Project on descriptive social networks of physicists
  - Data on correspondence with wider social circles – non-physicists
- Prussian Educ. Reforms in the early 19th c. and Human Capital Formation
  - A study on whether Prussian Reform Movement produced more prominent academics and individuals overall
  - Comparison of Prussia vs. other German states (state and city level)

Ongoing and Future Work

- Young vs older generations of physicists.
- Heterogeneity by the intensity of the pre-1925 correspondence.
- Additional network layers: co-authorship, co-citation, co-location.

1 Introduction

2 Historical setting

3 Data and Treatment Definition

4 Empirical Strategy and Results

5 Robustness and Heterogeneity

6 Conclusion

7 Appendix

History appendix

- The first consistent framework for understanding atomic phenomena.
- Why only eight founders? People may ask about this. Find evidence from historians on how quantum mechanics was being developed by only a narrow circle of physicists.
- Social circle of Bohr – Bohr's kindergarten (many young physicists influenced by him).
- The orthodox interpretation of QM – the Copenhagen interpretation – was coined in Copenhagen at Bohr's school.

History appendix: Scientific Communication after World War I

- New findings circulated through international journals, including *Zeitschrift für Physik* and *Physical Review*.
- Quick access to journals even over the Atlantic.
- Publishing process was similar to the modern one.
- Conferences, seminars, and the telephone created opportunities for more immediate scientific exchange.
- Private correspondence remained important for:
  - Sharing results and providing feedback
  - Discussing unpublished ideas
  - Maintaining international ties.

Letters therefore provide a rare record of informal scientific interaction.

History Appendix: Niels Bohr's (1885–1962) Career Episodes in Brief

- 1911 Completed his doctorate in physics at the University of Copenhagen
- 1912 Invited by Ernest Rutherford to work at Manchester University (1912–1916)
- 1913 Bohr's atomic model. Published his ideas in three articles in 1913
- 1916 Professor of theoretical physics at the University of Copenhagen
- 1920 Bohr established the Institute of Theoretical Physics
- "The Niels Bohr Institute" becomes the cutting edge frontier for modern physics
- Numerous foreign students and researchers visited Copenhagen

1922 Nobel Prize in Physics for "his services in the investigation of the structure of atoms and the radiation emanating from them".

History Appendix: The Clash of Titans

- Quantum jumps were supposition introduced by Bohr to explain the behavior of electrons.
- Bohr just postulated the jumps existed, but one couldn't visualize the jumps.
- There was no precedent for this. That seemed to go against the principles of science.
- Science was about understanding the world, not making up rules that seem to fit the data.
- Inevitable conflict between the two generations of scientists – older and younger ones.
- Einstein was an older generation scientist who fought with Bohr's ideas in the 1920s.

Go back to Bohr-Einstein comparison: Go back

Robustness Appendix

- Niels Bohr obtained the Nobel Prize in 1922. Delete correspondence in years 22-25.
- Focus on those who knew him before 1922:
  - Uni of Copenhagen (studying together)
  - Manchester Uni (exposure to Rutherford school) – but different trajectories after
- *https://en.wikipedia.org/wiki/Hans$_K$ramers*

- *Kramers could not choose Born as his supervisor due to the war.*
- *That is why he traveled to Niels Bohr.*
- *What about Sommerfeld, Ehrenfest, Slater, Kramers, Goudsmit, Uhlenbeck?*

Potential Mechanisms and Concerns Appendix

- A new generation of scientists. Bold, brilliant and above all young.
- It was crucial that they don't have attachment to ideas held by previous generations.
- Possibly they drive the rise in citations and publications? Check entering into profession.
- Niels Bohr and his "kindergarten" was one of the first of them.
- Concern: selection into elite scientific circles on early stage. Young scientists?
- Scientists reaching Bohr after the success of QM (post-1927)?

TWFE DiD: Nb of publications – In Levels

Back to Baseline results Go back

Period	1919–1932					
Outcome	Annual publication share			Cumulative publication share		
	(1)	(2)	(3)	(4)	(5)	(6)
Connected × Post-1925	0.012	-0.018	0.048	2.622***	2.391***	1.661***
	(0.061)	(0.067)	(0.069)	(0.517)	(0.551)	(0.547)
Specialty × Post		✓	✓		✓	✓
Country × Post		✓	✓		✓	✓
First publication year × Post		✓	✓		✓	✓
Birth year × Post		✓	✓		✓	✓
Network characteristics × Post			✓			✓
Scientist FE	✓	✓	✓	✓	✓	✓
Year FE	✓	✓	✓	✓	✓	✓
Observations	18,102	13,510	9,520	18,102	13,510	9,520
R^2	0.340	0.368	0.386	0.943	0.952	0.961

Notes: TWFE DiD regressions. Publication share adjusts each publication by the number of co-authors. Standard errors clustered at the scientist level.

$^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.10$.

List of WoS Journals Back

- ANNALEN DER PHYSIK 1900
- ANNALES DE PHYSIQUE 1946
- PHILOSOPHICAL MAGAZINE 1900
- PROCEEDINGS OF THE ROYAL SOCIETY OF LONDON SERIES A* 1905
- PHYSICAL REVIEW 1900
- USPEKHI FIZICHESKIKH NAUK 1955
- ZEITSCHRIFT FUR PHYSIK 1920
- PHYSIKALISCHE ZEITSCHRIFT 1900
- NATURWISSENSCHAFTEN 1913
- NATURE 1900
- SCIENCE 1900

Outcomes definition

Go back to Outcomes and other variables

- **Annual citations**: total citations received by all of a scientist's publications in year t.
- **Cumulative citations**: total citations received by all publications up to year t,

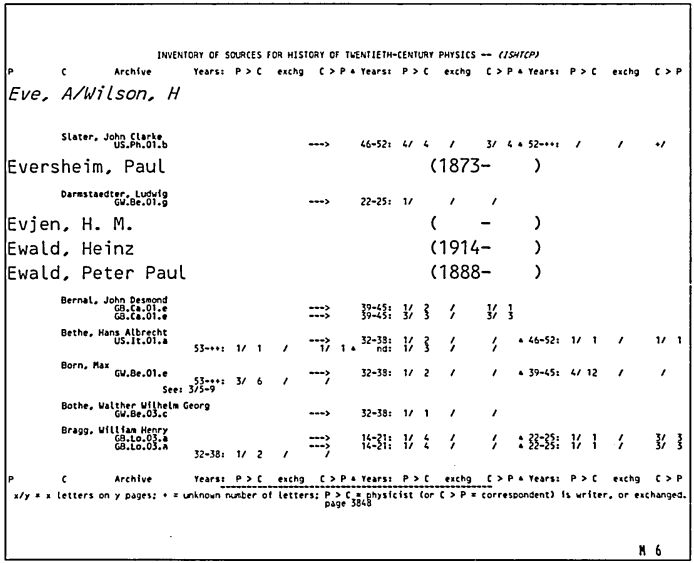
$$CumCites_{it} = \sum_{\tau \leq t} AnnualCites_{i\tau}.$$

- **Citations per article (CPA)**: cumulative citations relative to cumulative publications,

$$CPA_{it} = \frac{CumCites_{it}}{CumPublications_{it}}.$$

Example #1 of a microfilm card

Go back



INVENTORY OF SOURCES FOR HISTORY OF TWENTIETH-CENTURY PHYSICS -- *(ISHTCP)*

P	C	Archive	Years:	P > C	exchg	C > P	▲ Years:	P > C	exchg	C > P	▲ Years:	P > C	exchg	C > P
Eve, A/Wilson, H														
	Slater, John Clarke	US.Ph.01.b				--->	46-52:	4/ 4	/	3/ 4	▲ 52-***:	/	/	+/
Eversheim, Paul									(1873-)					
	Darmstaedter, Ludwig	GW.Be.01.g				--->	22-25:	1/	/	/				
Evjen, H. M.									(-)					
Ewald, Heinz									(1914-)					
Ewald, Peter Paul									(1888-)					
	Bernal, John Desmond	GB.Ca.01.e				--->	30-46:	1/ 2	/	1/ 1				
		GB.Ca.01.e				--->	39-45:	3/ 3	/	3/ 3				
	Bethe, Hans Albrecht	US.It.01.a	53-***:	1/ 1	/	---> 1/ 1	32-38: ▲ nd:	1/ 2 1/ 3	/ /	/ /	▲ 46-52:	1/ 1	/	1/ 1
	Born, Max	GW.Be.01.e	53-***:	3/ 6	/	---> /	32-38:	1/ 2	/	/	▲ 39-45:	4/ 12	/	/
		See: 3/5-9												
	Bothe, Walther Wilhelm Georg	GW.Be.03.c				--->	32-38:	1/ 1	/	/				
	Bragg, William Henry	GB.Lo.03.a				--->	14-21:	1/ 4	/	/	▲ 22-25:	1/ 1	/	3/ 3
		GB.Lo.03.A	32-38:	1/ 2	/	---> /	14-21:	1/ 4	/	/	▲ 22-25:	1/ 1	/	3/ 3

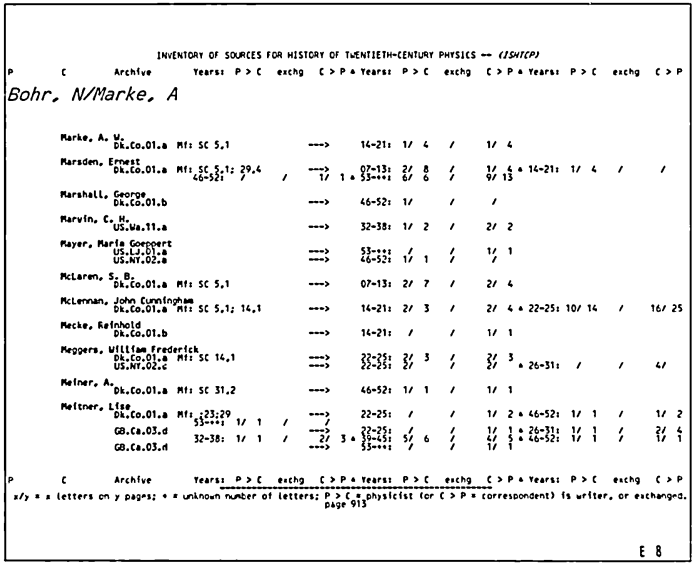
P	C	Archive	Years:	P > C	exchg	C > P	▲ Years:	P > C	exchg	C > P	▲ Years:	P > C	exchg	C > P

x/y = x letters on y pages; + = unknown number of letters; P > C = physicist (or C > P = correspondent) is writer, or exchanged.

M 6

Example #2 of a microfilm card

Go back



INVENTORY OF SOURCES FOR HISTORY OF TWENTIETH-CENTURY PHYSICS -- *(ISHTCP)*

P	C	Archive	Years:	P > C	exchg	C > P	Years:	P > C	exchg	C > P	Years:	P > C	exchg	C > P

Bohr, N/Marke, A

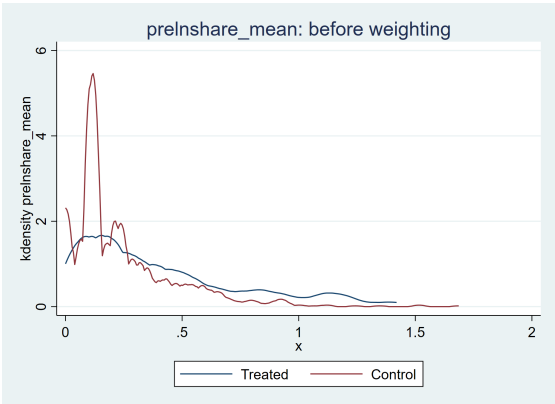
	Marke, A. W.	Dk.Co.01.a	Mf: SC 5,1		--->		14-21:	1/ 4	/	1/ 4				
	Marsden, Ernest	Dk.Co.01.a	Mf: SC 5,1: 29,4	/	--->		07-13:	2/ 8	/	1/ 4	a 14-21:	1/ 4	/	/
			46-52: /	/	1/	1 a	53-**:	6/ 6	/	9/ 13				
	Marshall, George	Dk.Co.01.b			--->		46-52:	1/	/	/				
	Marvin, C. H.	US.Wa.11.a			--->		32-38:	1/ 2	/	2/ 2				
	Mayer, Maria Goeppert	US.Ll.01.a			--->		53-**:	/	/	1/ 1				
		US.NY.02.a			--->		46-52:	1/ 1	/	/				
	McLaren, S. B.	Dk.Co.01.a	Mf: SC 5,1		--->		07-13:	2/ 7	/	2/ 4				
	McLennan, John Cunningham	Dk.Co.01.a	Mf: SC 5,1; 14,1		--->		14-21:	2/ 3	/	2/ 4	a 22-25:	10/ 14	/	16/ 25
	Mecke, Reinhold	Dk.Co.01.b			--->		14-21:	/	/	1/ 1				
	Meggers, William Frederick	Dk.Co.01.a	Mf: SC 14,1		--->		22-25:	2/ 3	/	2/ 3				
		US.NY.02.c			--->		22-25:	2/	/	2/	a 26-31:	/	/	4/
	Meiner, A.	Dk.Co.01.a	Mf: SC 31,2		--->		46-52:	1/ 1	/	1/ 1				
	Meitner, Lise	Dk.Co.01.a	Mf: 23;29		--->		22-25:	/	/	1/ 2	a 46-52:	1/ 1	/	1/ 2
			53-**: 1/ 1	/	/									
		GB.Ca.03.d			--->		22-25:	/	/	1/ 1	a 26-31:	1/ 1	/	2/ 4
			32-38: 1/ 1	/	2/	3 a	39-45:	5/ 6	/	4/ 5	a 46-52:	1/ 1	/	1/ 1
		GB.Ca.03.d			--->		53-**:	/	/	1/ 1				

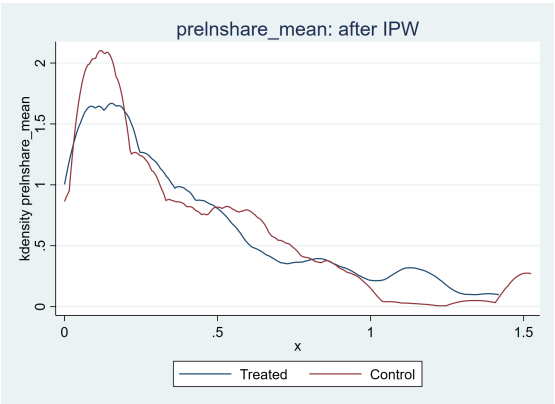
P	C	Archive	Years:	P > C	exchg	C > P	Years:	P > C	exchg	C > P	Years:	P > C	exchg	C > P

x/y = x letters on y pages; + = unknown number of letters; P > C = physicist (or C > P = correspondent) is writer, or exchanged.

E 8

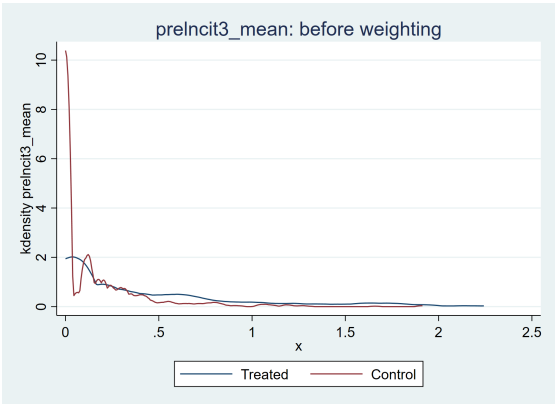
Distribution of pre-1925 Share of authors: before and after reweighting

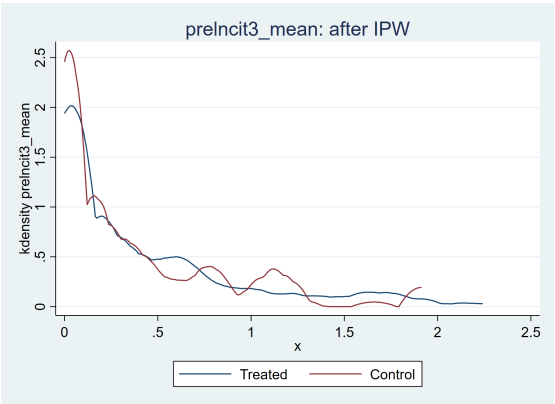




Go back

Distribution of pre-1925 Cites 3y: before and after reweighting





Go back

No-anticipation assumption

- Started off by Heisenberg (1925) and Schrödinger (1926) *independently* from each other
  - Unexpected *simultaneous* discovery
  - *Unprecedentedly fast* path of discovery – basic principles completed by 1927
- Two different theories about the same nature phenomena
  - Heisenberg's Matrix mechanics: anti-causal, probabilistic, discrete and with new formalism (infinite matrices)
  - Schrödinger's Wave mechanics: causal, deterministic, continuous and with old formalism (differential equations)
- Immediately accepted
  - Matrix Mechanics $\Rightarrow$ Göttingen-Munich-Copenhagen
  - The matrix mechanics, by the end of 1925, was recognized as an important advance throughout the German-speaking Central-European physicists and abroad, where was good relation with this Göttingen: Leiden, Cambridge-Engl., Cambridge-Mass., Pasadena (California)
  - Wave Mechanics $\Rightarrow$ other German-speaking physicists and countries outside of Europe
  - However, Schroedinger's theory was accepted enthusiastically among causalists. Outside of Germany it was adopted far more widely and enthusiastically than was the matrix mechanics.

Puzzling Contradictions Brought by QM

- Werner Heisenberg's described particle as particles, while Erwin Schrödinger's theory described particles as waves and many physicists travelled to Copenhagen to consult with each other over this puzzling contradiction.

Aside

"I, at any rate, am convinced that He is not playing at dice." (Einstein to Born, Dec'26)